Insertion Sort

```
void insertion_sort(int arr[], int size)
{
        for(int i=1; i<size; i++)
        {   int key=arr[i];  int j=i-1;
            while( arr[j]>key && j>=0)
            {
                    arr[j+1]=arr[j];   j--;
            }
            arr[j+1]=key;
        }
}
```

0, 1, 2, 3 —— 9

Properties:

i) Substract & Conquer ii) stable iii) internal (inplace)
iv) Data Dependant iv) Worst-case: $O(n^2)$, Best: $O(n)$

input array: 9, 2, 8, 1, 7, 3, 6, 5

Pass1: i=1, key=2, j=0 | pass 5: i=5, key=3, j=4,0

Step 1: j=0, 9,9,8,1,7,3,6.	1 2 7 € q q 6 s
array: 2,9,8,1,7,3,6,5	6te z: 0=3, 8,8,9,6,5
PasS2•o i-,2 key=8, j=1,0	ste 1,2,7,7,8,8 9 6
j=1, 2,9,9,1,7,3,6,5	step4: j=1 (Not executed)
2: j=0,	array: 1,2,3,7,8,9 6
array: 2,8,9 3 6	key=6, j=5,0
Pass 3: 3, key=1, j=2,0	ste 1 3 7 8, 9
te 2 2 g 01 q 736	step2: j=4, 2 3 7 7,8,8 9 5
step 2: j=1 , 28 g q 3	step3: j=3, 1,2,3,7,7,8
step 3: j= , 2 2 g 7	step4: j=2 (Not executed)
1,2,8,9, 7,3,6,5	1,2,3,6,7,8,9
i=4, key=7, 130	

3, 1,2,8,9,9,3,6,5	
1,2,8,8,9,3,6	
step3: j=1,	

{1,2,7,8,9}

array: 1, 2, 3, 6, 7, 8, 9, 5

pass 7: i = 7, key = 5, j = 6, 0

step 1: j = 6, 1, 2, 3, 6, 7, 8, 9, 9

step 2: j = 5, 1, 2, 3, 6, 7, 8, 8, 9

step 3: j = 4, 1, 2, 3, 6, 7, 7, 8, 9

step 4: j = 3, 1, 2, 3, 6, 6, 7, 8, 9

~~step 5: j = 2 (Not executed)~~

array: $\boxed{1, 2, 3, 5, 6, 7, 8, 9}$

5 → sort
→ 6 pass

8 7th pass

7

{5 * (A ~~[illegible]~~)^(~~[illegible]~~ (* B))}

input array: 9, 2, 8, 1, 7, 3, 6, 5

Pass 1:	2, 9, 8, 1, 7, 3, 6, 5	key = 2
Pass 2:	2, 8, 9, 1, 7, 3, 6, 5	key = 8
pass 3:	1, 2, 8, 9, 7, 3, 6, 5	key = 1
pass 4:	1, 2, 7, 8, 9, 3, 6, 5	key = 7
pass 5:	1, 2, 3, 7, 8, 9, 6, 5	key = 3
pass 6:	1, 2, 3, 6, 7, 8, 9, 5	key = 6
pass 7:	1, 2, 3, 5, 6, 7, 8, 9	key = 5

pass 8: